

Justin M. Parra

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EDUCATION

Rutgers University – New Brunswick

New Brunswick, NJ

Bachelor of Science in Electrical and Computer Engineering (ECE)

Expected December 2027

- **Relevant Coursework:** Principles of Electrical Engineering I & II, Microelectronic Circuits, Linear Systems and Signals, Control Systems Principles, Feedback & State Estimation, Digital Logic Design, Computer Architecture, Embedded Systems & Microcontrollers.

TECHNICAL SKILLS

Languages: C, C++ (C++11/17/20), Python, TypeScript, JavaScript, MATLAB, Verilog HDL, Assembly

Embedded & Microcontrollers: ESP32-S3, STM32F405, STM32, ARM Cortex-M4, Bare-Metal C, FreeRTOS, HAL, Timer ISRs

Hardware & PCB Design: Altium Designer, KiCad 8.0, 4-Layer PCB Routing, Controlled Impedance, Coplanar Ground Shielding, Analog Signal Conditioning, Active Op-Amp Filters, RLC Transient Analysis, TVS Surge Protection, Decoupling

Protocols & Buses: I2C (400 kHz), SPI, UART, CAN Bus, PWM Motor Control, ADC/DAC, RS-485, GPIO

Robotics & Control: Cascade PID Control, Differential Drive Kinematics, Sensor Fusion (EMA, EKF Principles), Modified

Flood Fill, A* Pathfinding, State Estimation

Tools & Platforms: Linux (Arch, Ubuntu, WSL2), Git, GitHub, CMake, LTspice, Make, GDB, Multisim

ENGINEERING EXPERIENCE

Rutgers Micromouse

New Brunswick, NJ

Lead Firmware & Hardware Engineer / Lead Controls Engineer / Co-President

September 2024 – Present

- Engineered real-time embedded **C/C++** firmware on **ESP32-S3** and **STM32** microcontrollers, utilizing **FreeRTOS** tasks to achieve deterministic **100 Hz** sensor polling and sub-millisecond motor control response.
- Architected **dual-loop cascade PID** feedback controllers fusing **100 Hz IMU** angular velocity (**Bosch BNO055 9-DOF**) with lateral distance error from **VL53L1X Time-of-Flight** sensors, eliminating motor drift and achieving sub-millimeter maze centering.
- Implemented memory-efficient **Modified Flood Fill** exploration and **A* pathfinding** algorithms in modern **C++** with bit-manipulated state matrices, reducing autonomous traversal solve time by **20%** under strict embedded SRAM limits.

LPT-KEYPAK

Tewksbury, NJ

Engineering Intern

June 2024 – August 2024

- Commissioned and validated automated high-speed packaging machinery for the **Bill & Melinda Gates Foundation**, boosting packaging line throughput by **30%** during pilot production qualification runs.
- Diagnosed electromechanical and process failure points across automated manufacturing equipment using structured root-cause analysis (**5-Whys**, **Pareto charts**), reducing production defect rates by **15%**.
- Interpreted **20+ industrial electrical schematics** to troubleshoot PLCs, motor drives, and sensor relays; authored technical documentation and maintenance SOPs to standardize workflows.

VEX Robotics (Rutgers IEEE)

New Brunswick, NJ

Lead Electrical Engineer

September 2024 – Present

- Architected communication and distributed power delivery architectures for competition robotics, engineering **RS-485 to UART to I2C** transceiver networks between VEX Brain and microcontrollers.
- Designed schematics and selected components including differential transceivers, TVS surge diodes, and bus termination resistors to protect sensitive logic from inductive spikes, reducing communication bus error rates to **<0.1%** across high-vibration competition runs.
- Executed hardware-in-the-loop (**HIL**) stress testing and signal integrity validation under severe mechanical vibration using digital oscilloscopes to verify packet-loss rates **<0.1%** and guarantee deterministic bus arbitration.

TECHNICAL PROJECTS

Autonomous Micromouse Robot (Jerrieee) - 2nd Place Regional Finalist

2025

C/C++, STM32F405RG, KiCad 8.0, 4-Layer ENIG PCB, DRV8847

- Won **2nd Place at Rowan Regional Competition**; designed custom **4-layer ENIG PCB** integrating **168 MHz ARM Cortex-M4** MCU, 5x optical ToF array over **400 kHz I2C**, and dual **DRV8847** motor drivers in an 84x76.5 mm chassis.
- Programmed real-time Flood-Fill maze navigation algorithm running in **sub-850 microseconds** per cell transition with a **1 kHz cascaded PID** control loop correcting gyro heading and velocity errors up to 4.2 m/s, implementing **anti-windup clamping** to maintain stable tracking.

Weave - Automated SPICE Netlist Parser & Schematic Compiler

2025

TypeScript, LTspice, ElkJS Graph Routing, Geometric Placement, EDA Tools

- Built an automated compiler converting raw SPICE subcircuit netlists (.cir) into formatted LTspice schematics (.asc) using hierarchical graph theory and layered placement algorithms.
- Implemented geometric Manhattan routing and net-label heuristics to eliminate overlapping wire segments and generate clear circuit schematics automatically.